

Journal Papers

1. Q. Yu, **D. He**, and H. Wang, "CA-ResNet: A Deep Learning Based Framework for Uplink Signal Detection of LoRa-Based LEO Satellite IoT," *IEEE Wireless Commn. Lett*, Early Access. 2026.
2. Q. Yu, **D. He**, H. Hou, and H. Wang, "Toward Future LoRa-Based LEO Satellite IoT: Opportunities, Framework, and Research Directions," *IEEE Commn. Stand. Mag.*, vol. 10, no. 2, pp. 191-198, Jun. 2026.
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4. T. Yang, **D. He**, W. Yuan, H. Liu, and H. Wang, "ISAC-Enabled Low-Altitude Economy: Game-Theoretic Learning Empowered Techniques and Future Directions," *IEEE Internet Things Mag.*, vol. 9, no. 3, pp. 48-54, May 2026.
5. N. Wu, G. Ding, Y. Fu, J. Zhang, and **D. He**, "Toward Space–Air–Ground Information Networks: Next Frontier for Wireless Communications," *Space Sci. Technol.*, 2025. (**Editorial**)
6. Y. Huang, H. Hou, **D. He**, G. K. Kurt and T. Q. S. Quek, "Guest Editorial Special Issue on Emerging Technologies Enhanced Cooperative Integrated Sensing and Communication in 6G Era," *IEEE Open J. Commun. Society*, vol. 6, pp. xiv-xvii, 2025. (**Editorial**)
7. J. Sun, **D. He**, H. Hou, P. Du, C. Fang, C. Zhang, W. Xu, D. Wang, and Y. Huang, "Cloud–Edge–End CF–MMIMO for UAV Swarms: Integrated Sensing and Robust Interference Detection in Perception Networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 13, pp. 3879–3893, 2026.
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10. Y. Huang, T. Mao, **D. He**, Q. Wang, C. Han, and Z. Wang, "Channel-Adaptive Autoencoder Architecture for Terahertz Aerospace Communications," *IEEE Trans. Veh. Technol.*, vol. 75, no. 2, pp. 3288-3293, Feb. 2026.
11. Q. Yu, D. Mishra, H. Wang, **D. He**, J. Yuan, and M. Matthaiou, "System-Level Performance Analysis of LoRa-Based LEO Satellite Constellations for IoT Communications," *IEEE Wireless Commun. Lett.*, vol. 14, no. 10, pp. 3279-3283, Oct. 2025.
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13. M. Yuan, **D. He**, H. Yin, H. Wang, F. Liu, Z. Wang, and T. Q.S. Quek, "Hybrid Beamforming for mmWave Integrated Sensing and Communication with Multi-static Cooperative

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14. Y. Chen, **D. He**, H. Wang, W. Yuan, and T. Q.S. Quek, "Joint Channel and Clipping Amplitude Estimation and Signal Detection for Clipped OTFS," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 335-360, 2026.
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 16. **D. He**, W. Yuan, J. Wu, and R. Liu, "Ubiquitous UAV Communication Enabled Low-Altitude Economy: Applications, Techniques, and 3GPP's Efforts," *IEEE Netw.*, vol. 40, no. 1, pp. 115-122, Jan. 2026.
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QPSK Detection Relying on Single-Bit Quantization," *IEEE Trans. Commun.*, vol. 69, no. 12, pp. 8175-8187, Dec. 2021.

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2. H. Zhao, **D. He**, H. Wang, and Y. Da, "Beampattern Gain Optimization for Near-Field Wideband Multi-User ISAC Systems," *2026 20th International Symposium on Wireless Communication Systems (ISWCS)*, Gold Coast, Australia, 2026, pp. 1-1.
3. Y. Zhang, T. Chan, and **D. He**, "VB-CPD-Based ISAC: Joint Tensor Rank and Parameter Estimation," *2026 20th International Symposium on Wireless Communication Systems (ISWCS)*, Gold Coast, Australia, 2026, pp. 1-1.
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8. M. Yuan, **D. He**, H. Yin and H. Wang, "Hybrid Beamforming for Millimeter-Wave ISAC System with Multi-Static Cooperative Localization," *2025 IEEE International Conference on Communications (ICC)*, Montreal, QC, Canada, 2025, pp. 6729-6734.
9. Q. Yu, D. Mishra, H. Wang, **D. He**, J. Yuan and M. Matthaiou, "Closed-Form Access Probability Analysis for LoRa-Based LEO Satellite IoT," *2025 IEEE International Conference on Communications (ICC)*, Montreal, QC, Canada, 2025, pp. 3193-3199.
10. Y. Chen, **D. He**, S. Li, and T. Mao, "Codebook Design for Holographic MIMO: Near-Field Prospects and Road to Standardization," *2025 International Wireless Communications and Mobile Computing (IWCMC)*, Abu Dhabi, United Arab Emirates, 2025, pp. 1084-1090. (**Best Paper Award**)
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13. L. Li, C. Yang, H. Li, and **D. He**, "DRL-Based Service Function Chains Embedding Through Network Function Virtualization in STINs," *2024 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, Zhuhai, China, 2024, pp. 1-5.
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15. T. Yang, **D. He**, H. Hou, H. Wang, Y. Huang, Z. Wang, and T. Q.S. Quek, "A Unified Tensor-Based Joint Communication and Sensing Parameter Estimation for ISAC with Large-Scale User Access," *2024 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, Zhuhai, China, 2024, pp. 1-6. (**Best Paper Award**)
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